The Failure Point of Systems of Linearized Differential Equations

How far from equilibrium does the linearization of a two-variable energy-balance climate model remain a valid predictor of the system's behaviour?

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Contents

1 Introduction **1**

2 Exploration **2**

1 Introduction

2 Exploration

With the aim of understanding how the number of rounds affect the orderedness of the cards in play, I experimented with collecting data on, and modelling the following relationship:

x-axis: the number of rounds played (independent variable) y-axis: the cards flipped from the draw pile (dependent variable)

I first used the variable of cards flipped from the draw pile because this is a key indicator that isolates the orderedness of only the cards in play rather than assessing the orderedness of the whole deck, reducing the scope and therefore increasing the validity. However, after collecting this data and graphing it on Desmos, I discovered that the relationship gave a random result that didn't indicate any kind of trend, like the one I had hypothesised.

Following this setback, I tried a different approach that still assessed the number of rounds as the independent variable, however changed the dependent variable to include the whole deck of cards, rather than just the few cards in play, faced up in front of each player. Additionally, I changed the actual determining factor to be the longest run that exists in each player's hand, rather than the number of cards drawn, to more accurately represent the orderedness of cards as it allows for cards in order, both up and down, to be recorded, rather than the indicator of cards drawn which can get skewed by the cards that players have available.

In this new exploration, I used the following factors:

Independent variable: number of the round Dependent variable: length of the longest run

(Be explicit about mistakes, new decisions, disregarding a certain approach etc.)